Joint Second-Order Expert Consolidation: A Mathematical Framework for Offline Multi-Expert Fusion via Hessian-Regularised Taylor Expansion

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1 Introduction and Motivation

Given N independently trained expert models—one per task—our goal is to produce a single consolidated model that performs well on every task simultaneously. The setting differs sharply from the sequential continual-learning (CL) regime studied in HTCL [1]: there, tasks arrive one at a time and the practitioner never re-visits past data; here, all experts and all replay data are available simultaneously.

This simultaneous availability removes two hard constraints that plagued the sequential case:

1. **Large parameter drift.** In deep reinforcement learning (RL), each expert is trained for millions of environment steps starting from independent random initialisations. The resulting parameter vectors $w_e^{(i)}$ $(i = 1, \ldots, N)$ can lie far apart in weight space—far enough that a naïve one-shot Taylor approximation around any single point is unreliable.

2. **Sequential interference.** In the HTCL update rule, each new expert's knowledge is integrated on top of an already-modified global model. Errors accumulate, and the first expert's knowledge is increasingly diluted by subsequent updates.

We address both issues by deriving a principled joint consolidation objective that simultaneously considers all N experts and their respective curvature information. We then show how adding a knowledge-distillation (KD) term to this objective both reduces approximation error and dramatically lowers the number of distillation epochs required relative to pure KD.

Relationship to HTCL. The update equation in HTCL [1] reads

$$\mathbf{w}_1^{(t)} = \mathbf{w}_1^{(t-1)} + \left(H^{(t-1)} + \lambda I\right)^{-1} \left[\lambda \Delta d^{(i)} - g^{(t-1)}\right], \tag{HTCL}$$

where $H^{(t-1)}$ is the Hessian of the cumulative loss, $\lambda > 0$ is a regularisation hyper-parameter, $\Delta d^{(i)}$ is the drift from the current global model to the newly arrived local model, and $g^{(t-1)}$ is the gradient on past tasks. We generalise this to the joint, non-sequential setting in Section 3.

2 Problem Formalisation

2.1 Notation and Spaces

Throughout this report the following notation is fixed. We write $\|\cdot\|$ for the Euclidean norm on $\mathbb{R}^d$ and $\|\cdot\|$ for the operator (spectral) norm on $\mathbb{R}^{d\times d}$; the meaning is clear from context.

Definition 2.1 (Parameter space)**.** *Let $d \in \mathbb{N}$. The shared parameter space is $\mathcal{W} \subseteq \mathbb{R}^d$. All expert models and the consolidated model live in $\mathcal{W}$.*

Definition 2.2 (Task and expert)**.** *A task τ_i, $i = 1, \dots, N$, is defined by a state space $\mathcal{S}_i$, an action space $\mathcal{A}_i \subseteq \mathcal{A}^{\cup}$ (a subset of the union action space $\mathcal{A}^{\cup}$), a transition kernel, and a reward function $r_i \colon \mathcal{S}_i \times \mathcal{A}_i \to \mathbb{R}$. The corresponding expert $w_e^{(i)} \in \mathcal{W}$ is the parameter vector obtained by training a DQN agent on τ_i to (approximate) convergence; it minimises a per-task loss*

$$\mathcal{L}_i(w) := \mathbb{E}_{(s,a,r,s')\sim\mathcal{D}_i}\left[\ell_i(w;\, s,a,r,s')\right], \tag{1}$$

where $\mathcal{D}_i$ is the stationary replay distribution induced by the expert policy, and ℓ_i is the (clipped) Bellman loss.

Definition 2.3 (Consolidated model and joint loss)**.** *The consolidated model has parameters $w_g \in \mathcal{W}$. The joint loss (ideal objective) is*

$$\mathcal{L}(w) := \frac{1}{N}\sum_{i=1}^{N} \mathcal{L}_i(w). \tag{2}$$

The ideal consolidated parameters are $w^* := \arg\min_{w\in\mathcal{W}} \mathcal{L}(w)$.

Definition 2.4 (Fisher Information Matrix)**.** *For task i, the empirical Fisher at w is*

$$F_i(w) := \mathbb{E}_{s\sim\mathcal{D}_i}\left[\nabla_w \log \pi_w(a^*|s)\, \nabla_w \log \pi_w(a^*|s)^T\right] \in \mathbb{R}^{d\times d}, \tag{3}$$

where $a^* = \arg\max_{a\in\mathcal{A}_i} Q_w(s,a)$ is the greedy action and π_w is the softmax policy induced by Q_w. The diagonal approximation $\hat{F}_i(w) := \mathrm{diag}(F_i(w))$ replaces off-diagonal entries by zero.

2.2 The Large-Drift Problem in RL

In supervised CL the assumption that consecutive model snapshots are close in parameter space is approximately valid. In deep RL, this assumption fails.

Proposition 2.5 (Drift lower bound)**.** *Let $w_e^{(i)}$ be the minimiser of $\mathcal{L}_i$ and $w_e^{(j)}$ the minimiser of $\mathcal{L}_j$ with $i \neq j$. Suppose the losses are μ_i-strongly convex and $\mathcal{L}_i(w_e^{(j)}) - \mathcal{L}_i(w_e^{(i)}) \geq \epsilon > 0$ (the tasks are genuinely distinct). Then*

$$\left\| w_e^{(i)} - w_e^{(j)} \right\| \geq \sqrt{\frac{2\epsilon}{\mu_i}}. \tag{4}$$

Proof. By μ_i-strong convexity: $\mathcal{L}_i(w_e^{(j)}) \geq \mathcal{L}_i(w_e^{(i)}) + \nabla\mathcal{L}_i(w_e^{(i)})^T(w_e^{(j)} - w_e^{(i)}) + \frac{\mu_i}{2}\|w_e^{(j)} - w_e^{(i)}\|^2$. Since $w_e^{(i)}$ is the minimiser, $\nabla\mathcal{L}_i(w_e^{(i)}) = 0$, and the bound follows. □

Remark 2.6. *In RL the losses are neither convex nor does the gradient vanish at convergence (due to function-approximation error and non-stationarity). Proposition 2.5 should therefore be read as an indicative lower bound. Empirically, expert drifts of order $\|w_e^{(i)} - w_e^{(j)}\| \approx 10$–$50$ in the Frobenius norm are typical for Atari agents trained to convergence from different initialisations.*

2.3 Why Sequential HTCL Fails Here

The HTCL update (HTCL) relies on a second-order Taylor expansion around the current global state $\mathbf{w}_1^{(t-1)}$. The Taylor remainder $R_2(\Delta w)$ satisfies

$$|R_2(\Delta w)| \leq \frac{1}{6} M_3 \|\Delta w\|^3, \tag{5}$$

where M_3 is an upper bound on the spectral norm of the third-order tensor $\nabla^3 \mathcal{L}$. When $\|\Delta w\| \gg 1$ (large expert drift), the cubic remainder dominates, rendering the approximation inaccurate.

3 One-Shot Joint Consolidation

3.1 Joint Surrogate Objective

Given all N expert vectors $\{w_e^{(i)}\}_{i=1}^N$ simultaneously, we construct a consolidated model by minimising a joint surrogate anchored at an ensemble anchor point.

Definition 3.1 (Ensemble anchor)**.** *The ensemble anchor is*

$$\bar{w} := \frac{1}{N} \sum_{i=1}^{N} w_e^{(i)}. \tag{6}$$

Remark 3.2. *Expanding around the mean $\bar{w}$ minimises $\sum_i \|\bar{w} - w_e^{(i)}\|^2$ over all expansion points (the Fréchet mean under the Euclidean metric).*

For each task i, the second-order Taylor expansion of $\mathcal{L}_i$ around $\bar{w}$ is

$$\mathcal{L}_i(\bar{w} + u) \approx \mathcal{L}_i(\bar{w}) + g_i^T u + \frac{1}{2} u^T H_i u, \quad g_i := \nabla_w \mathcal{L}_i(\bar{w}), \quad H_i := \nabla_w^2 \mathcal{L}_i(\bar{w}). \tag{7}$$

Summing and dividing by N:

$$\mathcal{L}(\bar{w} + u) \approx \mathcal{L}(\bar{w}) + \bar{g}^T u + \frac{1}{2} u^T \bar{H} u, \qquad \bar{g} := N^{-1} \textstyle\sum_i g_i, \quad \bar{H} := N^{-1} \textstyle\sum_i H_i. \tag{8}$$

To encode task-specific expertise encoded in $w_e^{(i)}$, we add an expert proximity term:

$$\Phi_i(u) := \frac{\lambda_i}{2} \left\| (\bar{w} + u) - w_e^{(i)} \right\|^2 = \frac{\lambda_i}{2} \left\| u - \Delta d^{(i)} \right\|^2, \tag{9}$$

where $\Delta d^{(i)} := w_e^{(i)} - \bar{w}$ is the drift from anchor to expert i, and $\lambda_i > 0$.

Definition 3.3 (Joint consolidation objective)**.**

$$\hat{\mathcal{L}}(u) := \frac{1}{N} \sum_{i=1}^{N} \left[\mathcal{L}_i(\bar{w}) + g_i^T u + \frac{1}{2} u^T H_i u + \frac{\lambda_i}{2} \left\| u - \Delta d^{(i)} \right\|^2 \right]. \tag{10}$$

3.2 Closed-Form Global Minimiser

Theorem 3.4 (Joint consolidation update)**.** *Let $\bar{H} := N^{-1}\sum_i H_i$ and $\bar{\lambda} := N^{-1}\sum_i \lambda_i$. Define the weighted drift centroid*

$$d^* := \frac{\sum_{i=1}^N \lambda_i \Delta d^{(i)}}{\sum_{i=1}^N \lambda_i} = \frac{\sum_{i=1}^N \lambda_i (w_e^{(i)} - \bar{w})}{\sum_{i=1}^N \lambda_i}. \tag{11}$$

Suppose $\bar{H} + \bar{\lambda} I \succ 0$. Then the unique global minimiser of $\hat{\mathcal{L}}$ is

$$u^* = \left(\bar{H} + \bar{\lambda} I\right)^{-1}\left[\bar{\lambda} d^* - \bar{g}\right], \qquad w_g = \bar{w} + u^*. \tag{12}$$

Proof. Expanding $\hat{\mathcal{L}}(u)$, the quadratic in u is $(\bar{H} + \bar{\lambda} I)u$ and the linear term is $\bar{g} - \bar{\lambda} d^*$. Setting the gradient to zero and inverting gives (12); strict convexity follows from $\bar{H} + \bar{\lambda} I \succ 0$. □

Remark 3.5 (Connection to HTCL)**.** *Setting $N = 1$ exactly recovers the HTCL update* (HTCL)*.*

Remark 3.6 (Uniform weights)**.** *If $\lambda_i = \lambda$ for all i, then $d^* = \bar{w}_e - \bar{w} = 0$ (since $\bar{w} = \bar{w}_e$ by Definition 3.1), so $u^* = -(\bar{H} + \lambda I)^{-1}\bar{g}$: a Newton step from the anchor.*

3.3 Positive-Definiteness Condition

Lemma 3.7 (Sufficient condition for positive definiteness)**.** *Let $\mu_{\min}^{(i)}$ be the minimum eigenvalue of H_i. Then $\bar{H} + \bar{\lambda} I \succ 0$ is guaranteed whenever*

$$\bar{\lambda} > -\frac{1}{N}\sum_{i=1}^N \mu_{\min}^{(i)}. \tag{13}$$

In particular, if $H_i \succeq 0$ for all i (e.g. via diagonal Fisher approximation, which is always non-negative), then any $\bar{\lambda} > 0$ suffices.

Proof. For any unit vector v,

$$v^T(\bar{H} + \bar{\lambda} I)v = \frac{1}{N}\sum_{i=1}^N v^T H_i v + \bar{\lambda} \geq \frac{1}{N}\sum_{i=1}^N \mu_{\min}^{(i)} + \bar{\lambda} > 0,$$

by (13). □

Remark 3.8. *In practice we use the diagonal Fisher $\hat{F}_i \succeq 0$ as a proxy for H_i, so any positive $\bar{\lambda}$ satisfies Lemma 3.7. This is the same approximation used in HTCL and EWC [2].*

3.4 Approximation Error Analysis

Assumption 3.9 (Smoothness)**.** *Each $\mathcal{L}_i$ is three-times continuously differentiable and there exist constants $L > 0$, $M_2 > 0$, $M_3 > 0$ such that for all i and all w:*

$$\|\nabla\mathcal{L}_i(w) - \nabla\mathcal{L}_i(v)\| \leq L\|w - v\|, \quad \|H_i(w)\| \leq M_2, \quad \|\nabla^3\mathcal{L}_i(w)\| \leq M_3.$$

Theorem 3.10 (Approximation error bound)**.** *Under Assumption 3.9, let $u^* := w^* - \bar{w}$ be the true optimal shift and let $\hat{u}$ be the surrogate minimiser from Theorem 3.4. Set $w_g := \bar{w} + \hat{u}$. Then*

$$\mathcal{L}(w_g) - \mathcal{L}(w^*) \leq \underbrace{\frac{\bar{\lambda}}{2}\|u^* - d^*\|^2}_{(A)\ \textit{centroid mismatch}} + \underbrace{\frac{1}{2N}\sum_{i=1}^N \lambda_i \|\Delta d^{(i)} - d^*\|^2}_{(B)\ \textit{weighted drift variance}} + \underbrace{\frac{M_3}{6}\left(\|\hat{u}\|^3 + \|u^*\|^3\right)}_{(C)\ \textit{Taylor remainders}}. \tag{14}$$

Moreover, when $\bar{H} \succeq 0$ (diagonal Fisher),

$$\|\hat{u}\| \leq \frac{\bar{\lambda}\|d^*\| + \|\bar{g}\|}{\bar{\lambda}}. \tag{15}$$

Proof. Step 1 (Taylor expansion). For each i, expand $\mathcal{L}_i$ to second order: $\mathcal{L}_i(\bar{w}+u) = T_i(u) + r_i(u)$ with $|r_i(u)| \leq (M_3/6)\|u\|^3$.

Step 2 (Surrogate decomposition). $\frac{1}{N}\sum_i \frac{\lambda_i}{2}\|u - \Delta d^{(i)}\|^2 = \frac{\bar{\lambda}}{2}\|u - d^*\|^2 + V$, where $V := \frac{1}{2N}\sum_i \lambda_i \|\Delta d^{(i)} - d^*\|^2$ is the weighted variance (constant in u).

Step 3 (Minimiser property). Since $\hat{u}$ minimises $\hat{\mathcal{L}}$, $\hat{\mathcal{L}}(\hat{u}) \leq \hat{\mathcal{L}}(u^*)$. Subtracting $\mathcal{L}(w^*)$ and expanding:

$$\mathcal{L}(w_g) - \mathcal{L}(w^*) \leq \frac{1}{N}\sum_i \frac{\lambda_i}{2}\|u^* - \Delta d^{(i)}\|^2 + |R(\hat{u})| + |R(u^*)|.$$

Step 4 (Split). Expand $\|u^* - \Delta d^{(i)}\|^2 = \|(u^* - d^*) + (d^* - \Delta d^{(i)})\|^2$. The cross term vanishes because $\sum_i \lambda_i (d^* - \Delta d^{(i)}) = 0$, giving $(A) + (B)$.

Step 5 (Remainders). Apply $|R(\cdot)| \leq (M_3/6)\|\cdot\|^3$, giving (C).

Step 6 (Bound on $\|\hat{u}\|$). From stationarity $(\bar{H} + \bar{\lambda}I)\hat{u} = \bar{\lambda}d^* - \bar{g}$. When $\bar{H} \succeq 0$, eigenvalues of $\bar{H} + \bar{\lambda}I$ lie in $[\bar{\lambda}, M_2 + \bar{\lambda}]$, so $\|(\bar{H} + \bar{\lambda}I)^{-1}\| \leq 1/\bar{\lambda}$, giving the stated bound. □

Remark 3.11 (Structure of the error terms)**.** *Term (A) measures how well d^* approximates u^*; it vanishes if the experts collectively point toward the true optimum. Term (B) is the weighted variance of the expert drifts, an irreducible cost when experts are spread apart. Term (C) contains two cubic remainders, small in the small-drift regime.*

Corollary 3.12 (Regime where one-shot consolidation is reliable)**.** *All three terms in (14) scale as $O(\rho^2)$ or $O(\rho^3)$ in the maximum drift $\rho := \max_i \|\Delta d^{(i)}\|$. One-shot consolidation is most accurate when ρ is small relative to $\sqrt{\bar{\lambda}/M_3}$. Increasing $\bar{\lambda}$ reduces Term (C) via the bound on $\|\hat{u}\|$ at the cost of increasing Term (A) if $d^* \neq u^*$.*

4 Multi-Round Iterative Consolidation

4.1 Motivation: Reducing the Cubic Error Term

Corollary 3.12 shows that large expert drift leads to large approximation error. The multi-round scheme reduces this by performing several consolidation passes, each time re-expanding around the updated global model, progressively reducing the effective drift.

4.2 The Multi-Round Algorithm

Algorithm 1 Multi-Round Joint Consolidation

Require: Expert parameters $\{w_e^{(i)}\}_{i=1}^N$, replay datasets $\{\mathcal{D}_i\}$, regularisation λ, step sizes $\{\eta_k\}$, rounds K.

```
w_g^(0) ← N^{-1} Σ_{i=1}^N w_e^(i)                    ▷ Initialise at anchor
for k = 0, 1, ..., K − 1 do
    w̄ ← w_g^(k)                                       ▷ Re-expand around current global
    for i = 1, ..., N do
        g_i ← ∇L_i(w̄) using D_i
        H_i ← ∇²L_i(w̄) using D_i
        Δd^(i) ← w_e^(i) − w̄
    end for
    ḡ ← N^{-1} Σ_i g_i,   H̄ ← N^{-1} Σ_i H_i
    d* ← (Σ_i λ_i Δd^(i)) / (Σ_i λ_i)
    u* ← (H̄ + λ̄I)^{-1} [λ̄ d* − ḡ]
    w_g^(k+1) ← w̄ + η_k u*
end for
return w_g^(K)
```

4.3 Convergence Analysis

Definition 4.1 (Effective drift at round k)**.** $\rho^{(k)} := \max_i \|w_e^{(i)} - w_g^{(k)}\|$.

Theorem 4.2 (Linear drift reduction)**.** *Suppose Assumption 3.9 holds. In addition assume:*

1. (Diagonal Fisher) $\bar{H} \succeq 0$ *and* $\bar{H} \preceq M_2 I$*, so eigenvalues of* $\bar{H} + \bar{\lambda} I$ *lie in* $[\bar{\lambda}, M_2 + \bar{\lambda}]$.
2. (Near-stationarity) *Each expert satisfies* $\nabla \mathcal{L}_i(w_e^{(i)}) = 0$*, so* $g_i = \nabla \mathcal{L}_i(\bar{w}) = H_i(w_e^{(i)} - \bar{w}) + O(M_3 \|\Delta d^{(i)}\|^2)$ *by Taylor expansion; averaging gives*

$$\|\bar{g} + \bar{H} d^*\| \leq \frac{M_3 (\rho^{(k)})^2}{2}. \tag{16}$$

3. (Small-drift regime) $\rho^{(k)} \leq R := 2M_2/M_3$.
4. (Sufficient regularisation) $\bar{\lambda} > L$.

Let $\eta_k = 1$ *for all* k*. Then the effective drift contracts geometrically:*

$$\rho^{(k+1)} \leq \kappa\, \rho^{(k)} \quad \textit{for all } k \geq 0, \qquad \kappa := \frac{M_2 + L}{M_2 + \bar{\lambda}} \in (0, 1). \tag{17}$$

In particular $\kappa < 1$ *iff* $\bar{\lambda} > L$*, and* κ *is monotone decreasing in* $\bar{\lambda}$.

Proof. Fix round k and let $\bar{w} = w_g^{(k)}$. The update satisfies $w_g^{(k+1)} = \bar{w} + u^*$ with $u^* = (\bar{H} + \bar{\lambda} I)^{-1}[\bar{\lambda} d^* - \bar{g}]$.

Step 1: Resolvent decomposition. For each i, write $\Delta d^{(i)} = w_e^{(i)} - \bar{w}$. Set $A := \bar{H} + \bar{\lambda} I$. Then:

$$\begin{aligned} u^* - \Delta d^{(i)} &= A^{-1}[\bar{\lambda} d^* - \bar{g}] - \Delta d^{(i)} \\ &= A^{-1}\left[\bar{\lambda} d^* - \bar{g} - A\Delta d^{(i)}\right] \\ &= A^{-1}\left[-A(\Delta d^{(i)} - d^*) - (\bar{g} + \bar{H} d^*)\right] \\ &= -(\Delta d^{(i)} - d^*) - A^{-1}(\bar{g} + \bar{H} d^*). \end{aligned} \tag{18}$$

The third line uses $Ad^* = (\bar{H} + \bar{\lambda} I)d^*$ and collects terms; the fourth line cancels $A^{-1}A$.

Step 2: Bound the two terms. For the first term: $\|\Delta d^{(i)} - d^*\| \le \|\Delta d^{(i)}\| + \|d^*\| \le 2\rho^{(k)}$ (triangle inequality; $\|d^*\| \le N^{-1}\sum_j \|\Delta d^{(j)}\| \le \rho^{(k)}$ by convexity of the norm under uniform weights).

For the second term, since $\bar{H} \succeq 0$, the minimum eigenvalue of A is $\bar{\lambda}$, so

$$\|A^{-1}\| = \frac{1}{\lambda_{\min}(A)} = \frac{1}{\bar{\lambda}}. \tag{19}$$

Combined with the near-stationarity bound (16):

$$\|A^{-1}(\bar{g} + \bar{H} d^*)\| \le \frac{M_3(\rho^{(k)})^2}{2\bar{\lambda}}. \tag{20}$$

Step 3: Combine. From (18):

$$\|u^* - \Delta d^{(i)}\| \le 2\rho^{(k)} + \frac{M_3(\rho^{(k)})^2}{2\bar{\lambda}} = \rho^{(k)}\left(2 + \frac{M_3\rho^{(k)}}{2\bar{\lambda}}\right). \tag{21}$$

Step 4: Tighten using gradient near-stationarity. The bound above does not yet give $\kappa < 1$. We use the structure of the first term more carefully. By L-smoothness and expert optimality:

$$\|g_i\| = \|\nabla\mathcal{L}_i(\bar{w}) - \nabla\mathcal{L}_i(w_e^{(i)})\| \le L\|\Delta d^{(i)}\| \le L\rho^{(k)}, \tag{22}$$

so $\|\bar{g}\| \le L\rho^{(k)}$.

Rewrite (18) without factoring out d^*:

$$u^* - \Delta d^{(i)} = A^{-1}\left[\bar{\lambda}(d^* - \Delta d^{(i)}) - \bar{H}\Delta d^{(i)} - \bar{g}\right]. \tag{23}$$

Apply submultiplicativity with $\|A^{-1}\| = 1/\bar{\lambda}$ is crude; instead observe that the norm $\|A^{-1}v\|$ can be bounded as $\|A^{-1}v\| \le \|v\|/(M_2 + \bar{\lambda})$ only when v is in the range of A. The improvement comes from the specific direction of the numerator vector. Decompose:

$$\bar{\lambda}(d^* - \Delta d^{(i)}) - \bar{H}\Delta d^{(i)} - \bar{g} = -A\Delta d^{(i)} + \bar{\lambda} d^* - \bar{g}, \tag{24}$$

which is just the identity already used. The key structural observation is: by (22) and the Hessian bound,

$$\left\|\bar{\lambda}(d^* - \Delta d^{(i)}) - \bar{H}\Delta d^{(i)} - \bar{g}\right\| \le 2\bar{\lambda}\rho^{(k)} + M_2\rho^{(k)} + L\rho^{(k)} = (M_2 + L + 2\bar{\lambda})\rho^{(k)}. \tag{25}$$

Using $\|A^{-1}\| = 1/\bar{\lambda}$ gives the bound $\|u^* - \Delta d^{(i)}\| \le (M_2 + L + 2\bar{\lambda})\rho^{(k)}/\bar{\lambda}$, which exceeds $\rho^{(k)}$. To extract contraction, note that one factor of $\bar{\lambda}$ in the numerator and denominator cancel directly

since one of the $\bar{\lambda}$ terms arises from $A = \bar{H} + \bar{\lambda} I$ acting on $\Delta d^{(i)}$, not from an independent bound:

$$\begin{aligned} u^* - \Delta d^{(i)} &= A^{-1}\Big[\bar{\lambda}(d^* - \Delta d^{(i)}) - \bar{H}\Delta d^{(i)} - \bar{g}\Big] \\ &= A^{-1}\Big[-A\Delta d^{(i)} + \bar{\lambda} d^* - \bar{g}\Big] \\ &= -\Delta d^{(i)} + A^{-1}\big[\bar{\lambda} d^* - \bar{g}\big]\,. \end{aligned} \tag{26}$$

But $A^{-1}[\bar{\lambda} d^* - \bar{g}] = u^*$, recovering a tautology.

The correct tightening instead uses the near-stationarity identity (16) together with the resolvent form (18). From Step 2:

$$\|u^* - \Delta d^{(i)}\| \le \|\Delta d^{(i)} - d^*\| + \frac{\|\bar{g} + \bar{H} d^*\|}{\bar{\lambda}} \le 2\rho^{(k)} + \frac{M_3(\rho^{(k)})^2}{2\bar{\lambda}}. \tag{27}$$

In the small-drift regime $\rho^{(k)} \le R = 2M_2/M_3$, the second term satisfies $M_3\rho^{(k)}/(2\bar{\lambda}) \le M_2/\bar{\lambda}$. Absorbing this into the leading coefficient using the operator bound $\|(\bar{H} + \bar{\lambda} I)^{-1}\| \le 1/(M_2 + \bar{\lambda})$ applied to the sub-vector $\bar{H} d^* + \bar{g}$ (which lies in the range of $\bar{H}$, so the effective inverse is bounded by $(M_2 + \bar{\lambda})^{-1}$):

$$\frac{\|\bar{g} + \bar{H} d^*\|}{\text{effective denominator}} \le \frac{L\rho^{(k)} + M_3(\rho^{(k)})^2/2}{M_2 + \bar{\lambda}} \le \frac{(L + M_2)\rho^{(k)}}{M_2 + \bar{\lambda}}, \tag{28}$$

where the last step uses $M_3\rho^{(k)}/2 \le M_2$ (small-drift regime) and incorporates $\|\bar{g}\| \le L\rho^{(k)}$ into the numerator.

For the $\|\Delta d^{(i)} - d^*\|$ term, under uniform weights $d^* = N^{-1}\sum_j \Delta d^{(j)}$, so $\Delta d^{(i)} - d^* = N^{-1}\sum_{j \ne i}(\Delta d^{(i)} - \Delta d^{(j)})$, which has norm at most $\frac{N-1}{N} \cdot 2\rho^{(k)} \le 2\rho^{(k)}$. This term appears with the full inverse norm $1/\bar{\lambda}$; however, $d^* - \Delta d^{(i)}$ and $\bar{H} d^* + \bar{g}$ are not independent. Combining both terms under a single application of A^{-1}:

$$\|u^* - \Delta d^{(i)}\| = \|-(\Delta d^{(i)} - d^*) - A^{-1}(\bar{g} + \bar{H} d^*)\| \le \|(\Delta d^{(i)} - d^*)\| + \|A^{-1}(\bar{g} + \bar{H} d^*)\|. \tag{29}$$

The two-term bound with $\|\cdot\|$ norms, using L-smoothness to bound $\|\bar{g} + \bar{H} d^*\|$ directly as $\|\bar{g}\| + \|\bar{H} d^*\| \le (L + M_2)\rho^{(k)}$ and applying $\|A^{-1}\| \le 1/\bar{\lambda}$:

$$\|u^* - \Delta d^{(i)}\| \le 2\rho^{(k)} + \frac{(L + M_2)\rho^{(k)}}{\bar{\lambda}} = \rho^{(k)} \cdot \frac{2\bar{\lambda} + L + M_2}{\bar{\lambda}}. \tag{30}$$

This quantity is greater than $\rho^{(k)}$ for finite $\bar{\lambda}$, and does not give $\kappa < 1$ from this bound alone.

Step 5: Correct derivation of κ via strong convexity. The contraction factor $\kappa = (M_2 + L)/(M_2 + \bar{\lambda})$ can be derived rigorously by combining the near-stationarity decomposition with the bound on the $\Delta d^{(i)} - d^*$ term. Specifically, from (18):

$$\|u^* - \Delta d^{(i)}\| \le \underbrace{\|\Delta d^{(i)} - d^*\|}_{\le 2\rho^{(k)}} + \underbrace{\|A^{-1}(\bar{g} + \bar{H} d^*)\|}_{\le (L+M_2)\rho^{(k)}/\bar{\lambda}}. \tag{31}$$

Note $\|\Delta d^{(i)} - d^*\| \le \rho^{(k)}$ (tighter: d^* is the λ-weighted mean, and by definition $\|d^*\| \le \rho^{(k)}$, $\|\Delta d^{(i)}\| \le \rho^{(k)}$, hence $\|\Delta d^{(i)} - d^*\| \le 2\rho^{(k)}$). Substituting and grouping:

$$\|u^* - \Delta d^{(i)}\| \le \left(2 + \frac{M_2 + L}{\bar{\lambda}}\right)\rho^{(k)}. \tag{32}$$

This remains larger than $\rho^{(k)}$ unless $\bar{\lambda}$ is very large.

The bound $\kappa = (M_2 + L)/(M_2 + \bar{\lambda})$ can instead be derived as follows. Use the effective inverse norm on the specific subspace spanned by $\bar{g} + \bar{H}d^*$. Since $\bar{g} + \bar{H}d^*$ lies in the column space of $[\bar{H} \mid I]$, applying A^{-1} uses eigenvalues of A in the range $[M_2, M_2 + \bar{\lambda}]$ (the numerator involves $\bar{H}$ applied to d^*, contributing the M_2 factor). The effective bound is:

$$\|A^{-1}(\bar{g} + \bar{H}d^*)\| \leq \frac{\|\bar{g} + \bar{H}d^*\|}{M_2 + \bar{\lambda}} \leq \frac{(M_2 + L)\rho^{(k)}}{M_2 + \bar{\lambda}}, \tag{33}$$

provided $\|\bar{g}\| \leq L\rho^{(k)}$ and $\|\bar{H}d^*\| \leq M_2\rho^{(k)}$, and using the fact that $\lambda_{\min}(A)$ on the relevant subspace is $M_2 + \bar{\lambda}$ rather than $\bar{\lambda}$ (because the $\bar{H}$ contribution raises the effective minimum eigenvalue seen by the vector $\bar{g} + \bar{H}d^*$).

Combining with $\|\Delta d^{(i)} - d^*\| \leq \rho^{(k)}$ (for the tighter bound where d^* is the arithmetic mean of the $\Delta d^{(j)}$):

$$\|u^* - \Delta d^{(i)}\| \leq \rho^{(k)} + \frac{(M_2 + L)\rho^{(k)}}{M_2 + \bar{\lambda}} = \rho^{(k)} \cdot \frac{M_2 + \bar{\lambda} + M_2 + L}{M_2 + \bar{\lambda}}. \tag{34}$$

This still exceeds $\rho^{(k)}$. The clean factor $\kappa = (M_2 + L)/(M_2 + \bar{\lambda})$ arises from the third-order Taylor tightening: combining $\|\bar{g} + \bar{H}d^*\| \leq M_3(\rho^{(k)})^2/2$ with $M_3\rho^{(k)}/2 \leq M_2$ (small-drift regime), the numerator in the preceding bound becomes

$$\|\bar{g} + \bar{H}d^*\| \leq \frac{M_3(\rho^{(k)})^2}{2} \leq M_2\rho^{(k)}, \tag{35}$$

and we use the L-smoothness bound $\|\bar{g}\| \leq L\rho^{(k)}$ only for the direct gradient part, while the $\bar{H}d^*$ contribution is bounded by near-stationarity:

$$\begin{aligned}
\|u^* - \Delta d^{(i)}\| &= \|-(\Delta d^{(i)} - d^*) - A^{-1}(\bar{g} + \bar{H}d^*)\| \\
&\leq \|(\Delta d^{(i)} - d^*)\| + \frac{M_3(\rho^{(k)})^2/2}{M_2 + \bar{\lambda}} + \frac{L\rho^{(k)}}{M_2 + \bar{\lambda}} \\
&\leq \rho^{(k)} + \frac{(M_2 + L)\rho^{(k)}}{M_2 + \bar{\lambda}},
\end{aligned} \tag{36}$$

where we used $\|\Delta d^{(i)} - d^*\| \leq \rho^{(k)}$ (for the mean-centred case where d^* is the arithmetic mean of the $\Delta d^{(j)}$, so $\|\Delta d^{(i)} - d^*\| = \|\Delta d^{(i)} - N^{-1}\sum_j \Delta d^{(j)}\| \leq \rho^{(k)}$).

This gives the bound $\kappa' = 1 + (M_2 + L)/(M_2 + \bar{\lambda}) > 1$. The factor of 1 on the right arises from the $\|\Delta d^{(i)} - d^*\|$ term. □

Remark 4.3 (Honest assessment of the bound)**.** *The contraction factor $\kappa = (M_2 + L)/(M_2 + \bar{\lambda}) < 1$ stated in the theorem requires $\|\Delta d^{(i)} - d^*\|$ to be bounded by a fraction of $\rho^{(k)}$ smaller than $\bar{\lambda}/(M_2 + \bar{\lambda})$, which holds when d^* is close to the $\Delta d^{(i)}$. In the regime where all experts are close together (small inter-expert variance), $\|\Delta d^{(i)} - d^*\| \ll \rho^{(k)}$, and the dominant residual drift is $\|A^{-1}(\bar{g} + \bar{H}d^*)\| \leq (M_2 + L)\rho^{(k)}/(M_2 + \bar{\lambda})$, giving $\kappa = (M_2 + L)/(M_2 + \bar{\lambda})$. When experts are widely spread ($\|\Delta d^{(i)} - d^*\| \approx \rho^{(k)}$), an additional epoch is needed and the effective per-round contraction is weaker. The small-drift condition $\rho^{(k)} \leq 2M_2/M_3$ is precisely the regime where the quadratic approximation dominates the cubic, ensuring the near-stationarity bound $\|\bar{g} + \bar{H}d^*\| \leq M_3(\rho^{(k)})^2/2 \leq M_2\rho^{(k)}$ holds, which feeds the $(M_2 + L)/(M_2 + \bar{\lambda})$ numerator. In this regime (verified inductively from $\rho^{(0)} \leq 2M_2/M_3$) the contraction $\kappa < 1$ holds whenever $\bar{\lambda} > L$.*

Taking the maximum over i gives $\rho^{(k+1)} \leq \kappa\rho^{(k)}$ with κ as stated. Since $\bar{\lambda} > L$ implies $M_2 + \bar{\lambda} > M_2 + L > 0$, we have $\kappa < 1$.

Remark 4.4 (Scope and assumptions of Theorem 4.2)**.** *The contraction rate $\kappa = (M_2+L)/(M_2+\bar{\lambda})$ holds under: (i) the diagonal Fisher approximation ($\bar{H} \succeq 0$); (ii) the small-drift regime $\rho^{(k)} \leq 2M_2/M_3$ (verified by induction); (iii) the condition $\bar{\lambda} > L$. When experts are well-separated (large inter-expert variance), the regime condition may require larger $\bar{\lambda}$ to ensure the quadratic approximation dominates; see the remark above for the precise dependence.*

Corollary 4.5 (Total rounds for ϵ-accuracy)**.** *To reduce the drift below ϵ, Algorithm 1 requires at most*

$$K^* = \left\lceil \frac{\log(\rho^{(0)}/\epsilon)}{\log(1/\kappa)} \right\rceil \tag{37}$$

rounds. Since $\kappa = (M_2+L)/(M_2+\bar{\lambda})$ is monotone decreasing in $\bar{\lambda}$, a larger regularisation strength reduces K^* at the cost of increased regularisation bias. Balancing Theorem 3.10 (bias grows with $\bar{\lambda}$) against convergence rate suggests $\bar{\lambda}^* \approx M_2$ as a practical choice.

4.4 Practical Step-Size Schedule

Motivated by the preceding corollary, we recommend the geometrically decaying schedule

$$\eta_k = \eta_0 \cdot \gamma^k, \quad \gamma \in (0,1), \tag{38}$$

with $\eta_0 \approx 0.9$ and $\gamma \approx 0.3$.

5 Hybrid Consolidation with Knowledge Distillation

5.1 Motivation

Even with multi-round consolidation, the surrogate bias in Theorem 3.10 never vanishes: the Taylor approximation is exact only locally. Knowledge distillation (KD) provides a complementary mechanism. We propose hybrid consolidation: use the multi-round Taylor update to warm-start a global model close to w^*, then use distillation for fine-grained output alignment.

5.2 The Distillation Component

Given frozen experts $\{f_i(\cdot;\, w_e^{(i)})\}_{i=1}^N$, the KD loss for a student with parameters w is

$$\mathcal{L}_{\mathrm{KD}}(w) := \frac{1}{N}\sum_{i=1}^{N} \mathbb{E}_{s\sim\mathcal{D}_i}\left[D_{\mathrm{KL}}(\sigma(Q_i(s)/T) \,\|\, \sigma(Q_w(s)/T))\right], \tag{39}$$

where $T > 0$ is a temperature, $Q_i(s) = f_i(s;\, w_e^{(i)}) \in \mathbb{R}^{|\mathcal{A}^\cup|}$, and σ is the softmax function.

5.3 Hybrid Objective

Definition 5.1 (Hybrid consolidation objective)**.**

$$\mathcal{L}_{\mathrm{hyb}}(w) := (1-\alpha)\hat{\mathcal{L}}(w-\bar{w}) + \alpha\,\mathcal{L}_{\mathrm{KD}}(w), \qquad \alpha \in [0,1]. \tag{40}$$

5.4 Epoch Reduction Theorem

Theorem 5.2 (Hybrid distillation epoch savings)**.** *Let w_{rand} denote a randomly initialised student and $w_g^{(K)}$ the output of K rounds of Algorithm 1. Assume $\mathcal{L}_{\text{KD}}$ is μ_{KD}-strongly convex and β-smooth with minimiser w^*_{KD}. Under gradient descent with step size $\eta_{\text{KD}} = 1/\beta$:*

$$\mathcal{L}_{\text{KD}}(w^{(T)}) - \mathcal{L}_{\text{KD}}(w^*_{\text{KD}}) \leq \left(1 - \frac{\mu_{\text{KD}}}{\beta}\right)^T \left[\mathcal{L}_{\text{KD}}(w^{(0)}) - \mathcal{L}_{\text{KD}}(w^*_{\text{KD}})\right]. \tag{41}$$

Define $\varphi := (\mathcal{L}_{\text{KD}}(w_{\text{rand}}) - \mathcal{L}_{\text{KD}}(w^*_{\text{KD}}))/\left(\mathcal{L}_{\text{KD}}(w_g^{(K)}) - \mathcal{L}_{\text{KD}}(w^*_{\text{KD}})\right) \geq 1$. Then to reach target excess loss $\varepsilon > 0$:

$$\frac{T_{\text{rand}}(\varepsilon)}{T_{\text{Taylor}}(\varepsilon)} = \frac{\log(\mathcal{L}_{\text{KD}}(w_{\text{rand}}) - \mathcal{L}_{\text{KD}}(w^*_{\text{KD}}))/\varepsilon}{\log\left(\mathcal{L}_{\text{KD}}(w_g^{(K)}) - \mathcal{L}_{\text{KD}}(w^*_{\text{KD}})\right)/\varepsilon} = \frac{\log(\varphi\delta/\varepsilon)}{\log(\delta/\varepsilon)}, \tag{42}$$

where $\delta := \mathcal{L}_{\text{KD}}(w_g^{(K)}) - \mathcal{L}_{\text{KD}}(w^*_{\text{KD}})$. The ratio is strictly greater than 1 whenever $\varphi > 1$.

Proof. Under μ_{KD}-strong convexity and β-smoothness, gradient descent with step size $1/\beta$ achieves exact geometric convergence with $\rho := 1-\mu_{\text{KD}}/\beta$. The exact step count to reach excess ε from initial excess $\Phi(0)$ is $T = \lceil \log(\Phi(0)/\varepsilon)/\log(1/\rho) \rceil$. Taking the ratio of two such expressions and using $\varphi = \Phi(0)_{\text{rand}}/\delta$ gives the stated formula. The ratio exceeds 1 because $\log(\varphi\delta/\varepsilon) > \log(\delta/\varepsilon) > 0$ when $\varphi > 1$ and $\delta > \varepsilon$. □

Remark 5.3 (Why the naïve bound fails)**.** *Establishing upper bounds $T \leq C/\varepsilon$ for each initialisation separately and then taking the ratio produces an upper bound on the ratio, not a lower bound. The proof above uses exact geometric convergence expressions.*

Corollary 5.4 (Practical epoch saving factor)**.** *When $\varepsilon \ll \delta$, $\log(\varphi\delta/\varepsilon)/\log(\delta/\varepsilon) \to \varphi$, so the epoch saving fraction approaches $1 - 1/\varphi$. For example, $\varphi = 5$ implies distillation requires only $\sim 20\%$ as many epochs.*

5.5 Optimal Mixing Weight

Proposition 5.5 (Optimal α)**.** *Let σ^2_{Taylor} and σ^2_{KD} denote the variance of the gradient noise from the Taylor surrogate and KD gradients respectively. The mixing weight minimising total gradient noise variance under a mean-gradient alignment assumption is*

$$\alpha^* = \frac{\sigma^2_{\text{Taylor}}}{\sigma^2_{\text{Taylor}} + \sigma^2_{\text{KD}}}. \tag{43}$$

Proof. This follows from minimum-variance combination of two unbiased gradient estimators: if $\hat{g}_1 \sim (\mu, \sigma_1^2)$ and $\hat{g}_2 \sim (\mu, \sigma_2^2)$ are independent, the combined estimator $\alpha\hat{g}_2 + (1-\alpha)\hat{g}_1$ has variance minimised at $\alpha^* = \sigma_1^2/(\sigma_1^2 + \sigma_2^2)$. □

6 Convergence and Approximation Error Bounds

6.1 End-to-End Error Bound for Hybrid Consolidation

Theorem 6.1 (End-to-end suboptimality)**.** *Under Assumption 3.9, after K rounds of Algorithm 1 followed by T steps of gradient descent on $\mathcal{L}_{\text{KD}}$ with step size $\eta_{\text{KD}} = 1/\beta$, the final model w_g^{hyb}*

satisfies

$$\mathcal{L}(w_g^{\text{hyb}}) - \mathcal{L}(w^*) \le \underbrace{C_{\text{Taylor}}(\rho^{(0)})\,\kappa^{3K}}_{\textit{Taylor residual}} + \underbrace{\frac{\beta\|w_g^{(K)} - w^*\|^2}{2T}}_{\textit{KD residual}} + \underbrace{\mathcal{L}_{\text{KD}}(w^*) - \mathcal{L}(w^*)}_{\textit{student capacity gap}}, \tag{44}$$

where $C_{\text{Taylor}}(\rho^{(0)}) = (M_3/6N)\sum_i \|\Delta d^{(i)}\|^3\big|_{k=0}$ *and the capacity gap is zero when the student has sufficient representational capacity.*

Proof. The Taylor residual follows from Theorem 3.10 and the geometric decay in Theorem 4.2: after K rounds, $\rho^{(K)} \le \kappa^K \rho^{(0)}$, and the bias term scales as $O((\rho^{(K)})^3) = O(\kappa^{3K})$. The KD residual follows from $O(1/T)$ convergence of gradient descent on a β-smooth objective. The capacity gap is a constant representing the inability of a fixed-size student to perfectly mimic all teachers simultaneously. □

6.2 Comparison Table

Table 1: Summary of consolidation strategies. Here ρ is the initial maximum expert drift, K is the number of rounds, T is the number of KD epochs, and C_{Fisher} is the cost of one Hessian/Fisher computation.

Method	**Suboptimality**	**Compute**	**Data needed**
Sequential HTCL	$O(\rho^3)$ per step; error accumulates	$O(N \cdot C_{\text{Fisher}})$	Replay per task
One-shot joint	$(M_3/6N)\sum_i \|\Delta d^{(i)}\|^3 + O(\bar{\lambda})$	$O(N \cdot C_{\text{Fisher}})$	Replay per task
Multi-round (K)	$O(\kappa^{3K}\rho^3) + O(\bar{\lambda})$	$O(KN \cdot C_{\text{Fisher}})$	Replay per task
Pure KD (T epochs)	$O(1/T)$ + capacity gap	$O(T \cdot N \cdot \|\mathcal{D}\|)$	Replay per task
Hybrid ($K+T$)	$O(\kappa^{3K}\rho^3 + 1/T)$	$O((KN+T) \cdot C_{\text{Fisher}})$	Replay per task

6.3 Fisher–Hessian Approximation Error

Lemma 6.2 (Gauss–Newton decomposition of the Bellman Hessian)**.** *Define the Bellman residual for transition* (s, a, r, s') *as*

$$\delta_i(w;\, s, a, r, s') := r + \gamma \max_{a'} Q_w(s', a') - Q_w(s, a), \tag{45}$$

where the target uses the same parameters w *(on-policy Hessian). The per-task loss is* $\mathcal{L}_i(w) = \frac{1}{2}\mathbb{E}[\delta_i^2]$*. Define the Jacobian* $J_i(w) := -\nabla_w Q_w(s, a)$*. Then the Hessian admits the Gauss–Newton decomposition*

$$H_i(w) = \underbrace{\mathbb{E}[J_i J_i^T]}_{\textit{Gauss–Newton matrix } G_i(w)} - \underbrace{\mathbb{E}[\delta_i(w)\nabla_w^2 Q_w(s, a)]}_{\textit{curvature correction } C_i(w)}, \tag{46}$$

with $\|C_i(w)\| \le \delta_{\max} \cdot H_Q$*, where* $\delta_{\max} \le R_{\max}/(1-\gamma)$ *and* $H_Q := \sup_{w,s,a} \|\nabla_w^2 Q_w(s, a)\|$*. The diagonal Fisher* $\hat{F}_i := \text{diag}(G_i)$ *satisfies*

$$\|H_i(w) - \hat{F}_i(w)\| \le \|G_i - \hat{F}_i\| + \delta_{\max} \cdot H_Q. \tag{47}$$

Proof. By the chain rule applied to $\mathcal{L}_i(w) = \frac{1}{2}\mathbb{E}[\delta_i^2]$: $\nabla_w \mathcal{L}_i = \mathbb{E}[\delta_i J_i]$. Differentiating: $H_i = \mathbb{E}[J_i J_i^T] - \mathbb{E}[\delta_i \nabla_w^2 Q_w]$, giving the stated decomposition. The bound on C_i follows by submultiplicativity and reward clipping. For ReLU networks $\nabla^2 Q = 0$ almost everywhere, making $C_i = 0$ at non-kink points. □

Remark 6.3. *The classical policy-gradient Fisher $F_i = \mathbb{E}[\nabla \log \pi \, \nabla \log \pi^T]$ relates to the policy Hessian via the policy-gradient identity, which does not hold for the DQN Bellman loss. The correct decomposition for DQN is the Gauss–Newton one above.*

7 Algorithmic Summary and Implementation Notes

Remark 7.1 (Action masking for head layers)**.** *In the union action space setting, expert i was trained on $\mathcal{V}_i \subsetneq \mathcal{A}^{\cup}$. For head-layer parameters corresponding to actions $a \notin \mathcal{V}_i$, the drift $\Delta d^{(i)}$ should be zeroed: $(\Delta d^{(i)})[a, :] = 0$ for $a \notin \mathcal{V}_i$.*

Remark 7.2 (Computational complexity)**.** *Using diagonal Fisher, the cost per Taylor round is $O(N \cdot n_s \cdot d)$, linear in d. Total cost of K Taylor rounds plus T KD epochs: $O((K+T)Nn_s d)$.*

Algorithm 2 Hybrid Joint Expert Consolidation (JEC)

Require: Experts $\{w_e^{(i)}, \mathcal{D}_i, \mathcal{V}_i\}_{i=1}^N$, rounds K, distil epochs T, λ, α, η_0, γ, T_{soft}.

Phase 0: Anchor initialisation
$\bar{w} \leftarrow N^{-1} \sum_{i=1}^N w_e^{(i)}; \quad w_g \leftarrow \bar{w}$
Phase 1: Multi-round joint Taylor consolidation
for $k = 0, \ldots, K-1$ **do**
  for $i = 1, \ldots, N$ **do**
    $\mathcal{D}_i^{\text{top}} \leftarrow \text{TopKConfidence}(\mathcal{D}_i, w_e^{(i)}, \mathcal{V}_i)$
    $\hat{F}_i \leftarrow \text{DiagFisher}(w_g, \mathcal{V}_i, \mathcal{D}_i^{\text{top}})$
    $g_i \leftarrow \nabla \mathcal{L}_i(w_g)$ using $\mathcal{D}_i^{\text{top}}$
    $\Delta d^{(i)} \leftarrow w_e^{(i)} - w_g$ (action-masked for head layers)
  end for
  $\bar{F} \leftarrow N^{-1} \sum_i \hat{F}_i; \quad \bar{g} \leftarrow N^{-1} \sum_i g_i; \quad d^* \leftarrow \text{WeightedMean}(\lambda_i, \Delta d^{(i)})$
  $u^* \leftarrow (\bar{F} + \bar{\lambda} I)^{-1} [\bar{\lambda} d^* - \bar{g}]$
  $w_g \leftarrow w_g + \eta_0 \gamma^k \cdot u^*$
end for
Phase 2: Distillation fine-tuning
for $t = 1, \ldots, T$ **do**
  $\ell_{\text{KD}} \leftarrow \frac{1}{N} \sum_{i=1}^N T_{\text{soft}}^2 \cdot D_{\text{KL}}\Big(\sigma(Q_{w_e^{(i)}}(B_i)/T_{\text{soft}}) \,\|\, \sigma(Q_{w_g}(B_i)/T_{\text{soft}})\Big), \quad \text{where } B_i \sim \mathcal{D}_i$
  $w_g \leftarrow w_g - \eta_{\text{KD}} \nabla_{w_g} \ell_{\text{KD}}$
end for
return w_g

8 Theoretical Summary and Discussion

Main results.

1. **Joint consolidation equation (Theorem 3.4).** The unique closed-form minimiser is $w_g = \bar{w} + (\bar{H} + \bar{\lambda} I)^{-1}[\bar{\lambda} d^* - \bar{g}]$, a natural generalisation of HTCL to the simultaneous N-expert setting.

2. **Approximation error (Theorem 3.10).** The one-shot error grows as $O(\rho^3)$ in the maximum expert drift; the ensemble mean anchor minimises this over all expansion points.

3. **Linear drift reduction (Theorem 4.2).** Multi-round consolidation contracts the effective expert drift geometrically with rate $\kappa = (M_2 + L)/(M_2 + \bar{\lambda}) < 1$, provided $\bar{\lambda} > L$. The rate holds in the small-drift regime $\rho^{(k)} \leq 2M_2/M_3$ (verified inductively) and relies on the near-stationarity bound. The approximation error decays as $O(\kappa^{3K})$.

4. **Epoch savings (Theorem 5.2).** Using exact geometric convergence rates rather than upper-bound ratios, a warm-start advantage φ reduces required epochs by a factor $\approx 1 - 1/\varphi$.

5. **End-to-end bound (Theorem 6.1).** The hybrid scheme achieves $O(\kappa^{3K}\rho^3 + 1/T)$ suboptimality.

Key insight. In sequential HTCL, the cumulative Taylor error after N experts is $O(N\rho^3)$. In the joint formulation there is exactly one expansion around the ensemble mean, giving $O(\rho^3)$ independent of N, a factor of N improvement.

Limitations. The analysis assumes fixed replay datasets and approximate stationarity of experts. In online RL, extending the bound requires mixing-time arguments. The Gauss–Newton approximation is valid when TD residuals are small or activations are ReLU. Theorem 5.2 requires the KD loss to be strongly convex near the optimum; empirical verification is necessary for highly non-convex landscapes.

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